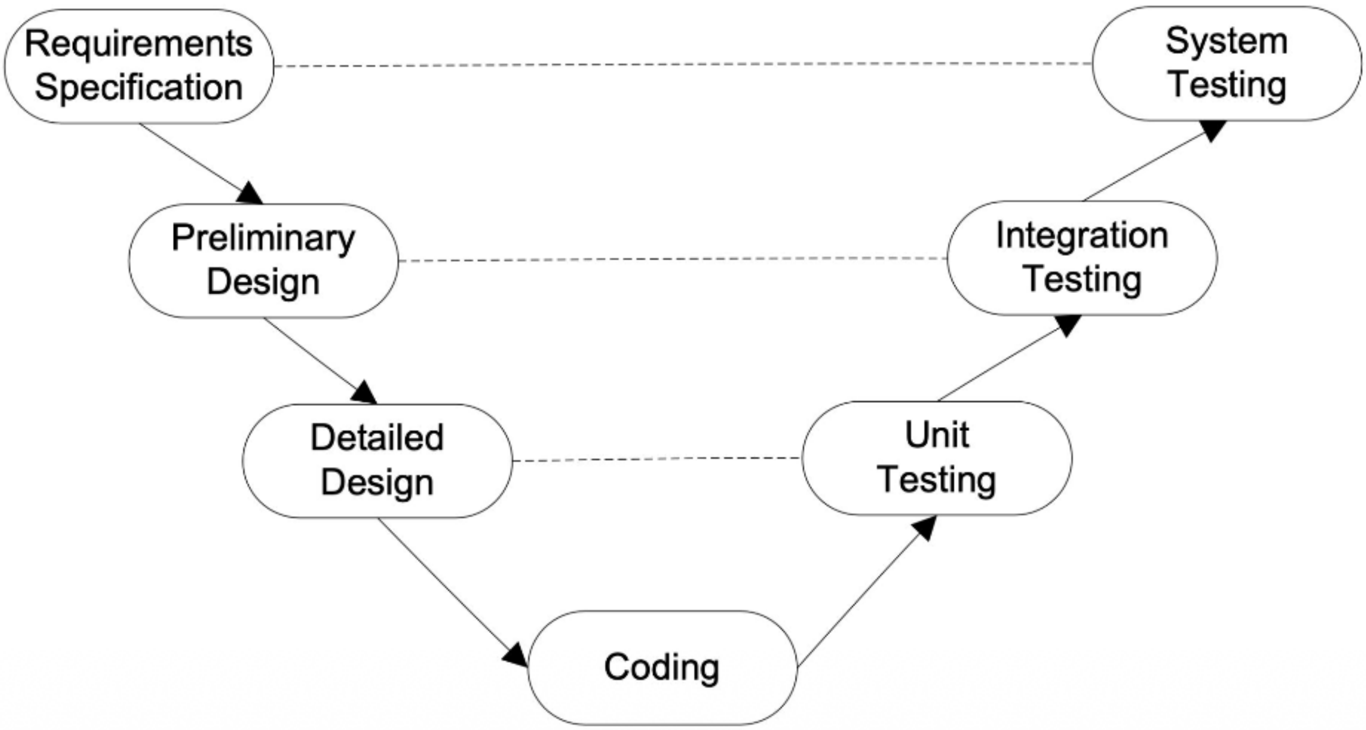
V Model, Waterfall Model

What is the V-Model?



The **V-Model** is a highly disciplined SDLC model where development and testing happen in parallel. It is also known as the **Verification and Validation (V&V) Model**, where every development phase has a corresponding testing phase.

Phases of the V-Model

1. **Verification Phase (Left Side)**

- **Requirements Specification**: Gathering user needs (linked to System Testing).
- **Preliminary Design**: Planning the high-level system architecture (linked to Integration Testing).
- **Detailed Design**: Designing the internal logic of individual modules (linked to Unit Testing).

2. **Coding (The Bottom)**

- **Coding**: The actual implementation where designs are converted into source code

3. **Validation Phase (Right Side)**

- **Unit Testing**: Testing individual modules or pieces of code.
- **Integration Testing**: Verifying that different modules work together correctly.
- **System Testing**: Testing the entire application against the original requirements.

Waterfall Model

The **Waterfall Model** is a **step-by-step** process where each phase must be finished before the next one starts. Like a real waterfall, the progress flows in only **one direction** and phases never overlap.

Key Characteristics

1. **Rigid:** You cannot go back to a previous stage
2. **Documentation-Heavy:** Every stage requires a formal document.
3. **High Risk:** The client doesn't see the product until the very end, which can lead to "Absence of error fallacy" if the requirements were misunderstood.

Phases of Waterfall Model

Requirement Analysis: Gathering and documenting all user needs.
Global Design: Planning the high-level system architecture.
Detailed Design: Defining internal logic and module structures.
Coding: Translating designs into actual source code.
Testing: Verifying the software to find and fix defects.
Deployment: Releasing the final product to the production environment.
Maintenance: Providing updates and fixing bugs after release.